Maths for Computing — Phase 1: Numbers, Binary, Modular Arithmetic and Sequences

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What this phase covers

This phase is all of Lecture 1 and Seminar 1. After it you can answer mock **Question 6** (IP addresses), **Question 22** (binary) and **Question 23** (encryption), plus revision Tasks 1, 2 and 3.

You need from Phase 0: remainders, negative numbers, powers of 2, and solving simple equations.

The big ideas, in order:

1. How numbers are written: decimal (base 10) and binary (base 2).
2. The families of numbers and their symbols.
3. Prime numbers and breaking numbers into primes.
4. The division theorem: dividing with a remainder, written as a formula.
5. Modular arithmetic: arithmetic that wraps around, like a clock.
6. Using modular arithmetic to encrypt words.
7. Sequences: lists of numbers that follow a rule.

1. Number systems: decimal and binary

Decimal (base 10)

The **decimal system**, also called **base 10**, is the normal way we write numbers. It uses ten digits, 0 to 9.

Each position is worth a **power of 10**. Moving one place left multiplies the value by 10.

Position value	$10^3 = 1000$	$10^2 = 100$	$10^1 = 10$	$10^0 = 1$
Digit in 3471	3	4	7	1
Worth	3000	400	70	1

So 3471 = 3000 + 400 + 70 + 1. You already do this without thinking. Binary works the same way with 2 instead of 10.

Binary (base 2)

The **binary system**, or **base 2**, uses only two digits: **0 and 1**.

Why do computers use it? A computer is made of tiny electrical switches. Each switch has two states: **off (0)** and **on (1)**. So every piece of data (numbers, text, images) is stored as 0s and 1s.

- One binary digit is called a **bit** (short for binary digit).
- **8 bits** make a **byte**.

To show a number is binary, we sometimes write a small 2 after it: 1110_2 means "1110 in binary". Without it, 1110 could be confused with one thousand one hundred and ten.

Binary place values

Each position is worth a **power of 2**, starting with $2^0 = 1$ on the right.

Position (from right, starting at 0)	7	6	5	4	3	2	1	0
Value	128	64	32	16	8	4	2	1

Binary to decimal

Method: write the place values above the digits. Add up the values where there is a 1. Ignore the 0s.

Example 1: 10011_2 (from your lecture)

Value	16	8	4	2	1
Digit	1	0	0	1	1

Add the values under the 1s: 16 + 2 + 1 = **19**.

The lecture writes the same thing as: $1\times2^4 + 0\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 = 16 + 0 + 0 + 2 + 1 = 19$.

Example 2: 11010_2 (Seminar 1)

Value	16	8	4	2	1
Digit	1	1	0	1	0

16 + 8 + 2 = **26**.

Example 3: 1001101_2

Value	64	32	16	8	4	2	1
Digit	1	0	0	1	1	0	1

64 + 8 + 4 + 1 = **77**.

Tip: always start labelling from the **right** with 1, then 2, 4, 8 … going left. Count the digits first so you know where to start.

Decimal to binary: method 1, repeated division by 2 (the lecture method)

1. Divide the number by 2. Write the quotient and the remainder (always 0 or 1).
2. Divide the quotient by 2 again. Keep going until the quotient is 0.
3. Read the remainders from **bottom to top**. That is the binary number.

Example: 19 to binary

Division	Quotient	Remainder
19 ÷ 2	9	1
9 ÷ 2	4	1
4 ÷ 2	2	0
2 ÷ 2	1	0
1 ÷ 2	0	1

Read the remainders upwards: 1, 0, 0, 1, 1. So 19 = $\mathbf{10011_2}$.

Example: 27 to binary

Division	Quotient	Remainder
27 ÷ 2	13	1
13 ÷ 2	6	1
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

Upwards: **11011_2**. Check: 16 + 8 + 2 + 1 = 27. Correct.

Decimal to binary: method 2, subtract powers of 2 (faster in an exam)

1. Find the biggest power of 2 that fits into the number. Put a 1 in that position.
2. Subtract it. Repeat with what is left.
3. Every position you did not use gets a 0.

Example: 45 to binary (mock Question 22)

- Powers of 2: 1, 2, 4, 8, 16, 32, 64. The biggest that fits in 45 is 32. 45 − 32 = 13.
- Biggest fitting 13 is 8. 13 − 8 = 5.
- Biggest fitting 5 is 4. 5 − 4 = 1.
- 1 fits 1. 1 − 1 = 0. Done.

Used: 32, 8, 4, 1. Not used: 16, 2.

Value	32	16	8	4	2	1
Digit	1	0	1	1	0	1

45 = **101101_2**. That is answer A in your mock.

Exam shortcuts for multiple choice

- **Odd numbers end in 1, even numbers end in 0** in binary. 45 is odd, so options ending in 0 (like 101110) are wrong instantly.
- **Count the digits.** A number from 32 to 63 has exactly 6 binary digits; 64 to 127 has 7. 45 needs 6 digits, so 1101101 (7 digits) is wrong.
- Convert each remaining option back to decimal to check.

Useful table (lecture)

Decimal	1	2	3	4	5	6	7	8	9	10
Binary	1	10	11	100	101	110	111	1000	1001	1010

Mini tasks 1

1. Convert 53 to binary.
2. Convert 102 to binary.
3. Convert 111000_2 to decimal.
4. Convert 11111_2 to decimal.
5. Which is 38 in binary: (A) 100110 (B) 100111 (C) 110010 (D) 1001100?

2. Types of numbers

Numbers are grouped into families. Each family has a name and a special letter, written in a double-struck style ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$). Mock Question 10 uses $\mathbb{Z}$, and seminar tasks use $\mathbb{N}$ and $\mathbb{R}$.

Family	Symbol	What's in it	Examples
Natural numbers	$\mathbb{N}$	Counting numbers	1, 2, 3, 4, …
Whole numbers	W	Natural numbers plus 0	0, 1, 2, 3, …
Integers	$\mathbb{Z}$	Whole numbers and their negatives	…, −2, −1, 0, 1, 2, …
Rational numbers	$\mathbb{Q}$	Anything that can be written as a fraction of two integers (bottom not 0)	½, ¾, −7/3, 5 (= 5/1), 0.25, 0.333…
Irrational numbers	I	Numbers that can **not** be written as a fraction	√2, √3, π, e
Real numbers	$\mathbb{R}$	All rationals and irrationals together: every point on the number line	3.14, −0.01, √2, 7

Family	Symbol	What's in it	Examples
Complex numbers	$\mathbb{C}$	Real numbers plus an imaginary part (just know the name)	2 + 3i

Where the letters come from: $\mathbb{Z}$ comes from the German word *Zahlen*, meaning "numbers". $\mathbb{Q}$ stands for **quotient**, because rationals are quotients (divisions).

Watch out: is 0 a natural number?

Your lectures disagree. Lecture 1 says $\mathbb{N} = \{1, 2, 3, ...\}$. Lecture 2 says $\mathbb{N} = \{0, 1, 2, ...\}$. **Always read what the question says.** For example, Seminar 2 Task 4.1 says "Let $\mathbb{N}$ be the set of natural numbers $\{1, 2, 3, ...\}$", so there 0 is not included. If a question does not say, Lecture 1's version (starting at 1) is the safer guess.

The families fit inside each other

Every natural number is a whole number. Every whole number is an integer. Every integer is rational. Every rational is real. Irrationals are real but not rational.

So: $\mathbb{N} \subseteq W \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$, and $I \subseteq \mathbb{R}$. (Read $\subseteq$ as "is inside" for now; Phase 2 explains it.)

How to classify a number

Ask these questions in order:

1. Can it be written without a fraction or decimal part? If yes and it is positive, it is natural. If it could be 0 or negative, it is an integer.
2. Can it be written as a fraction a/b with integers? Then it is rational. Decimals that stop (0.75) or repeat forever (0.333...) are rational.
3. Is it a square root that does not come out exactly, or π? Then it is irrational. Its decimals go on forever with no repeating pattern.

Examples:

- −7: integer (also rational, also real). Not natural.
- 0.75 = 3/4: rational.
- $\sqrt{16} = 4$: natural! Don't be fooled by the $\sqrt{}$ sign; simplify first.
- $\sqrt{5} \approx 2.236...$: irrational.
- −½: rational, not an integer.

Programming equivalents (from the lecture)

Maths set	Typical data type	Note
$\mathbb{N}$ and $\mathbb{Z}$	int	Computers store both the same way, in fixed sizes (32-bit, 64-bit)
$\mathbb{Q}$	Fraction (Python) or two ints	Usually stored as numerator and denominator
$\mathbb{R}$	float or double	Only approximated, because memory is finite
$\mathbb{C}$	complex (Python), Complex class	Not built in to every language

Why integers are popular in code: they are fast for the processor, exact (no rounding errors), use little memory, are used for loop counters and array positions (indexes), and work with bit-level operations.

Mini tasks 2

1. Name the smallest family each belongs to: 12, −3, 2/5, $\sqrt{7}$, $\sqrt{49}$, 0.
2. True or false: every integer is a rational number.
3. True or false: $\sqrt{2}$ is a rational number.
4. Which data type would a programmer usually use for 3.14?

3. Prime numbers and prime factorisation

Number theory is the part of maths that studies integers and their properties. Primes are its building blocks.

Divisors (factors)

A **divisor** or **factor** of a number divides it exactly, with remainder 0. The divisors of 12 are 1, 2, 3, 4, 6, 12.

Prime and composite

- A **prime number** is a natural number **greater than 1** whose only divisors are 1 and itself. Example: 5 (only 1 and 5 divide it).

- A **composite number** is a natural number greater than 1 that is **not** prime: it has some other divisor. Example: 4 (divisible by 1, 2 and 4).
- **1 is neither** prime nor composite. This is a common trick question.
- **2 is the only even prime.** Every other even number is divisible by 2.

Primes up to 50 (learn them): **2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47**.

Quick divisibility tests

Divisible by	Test	Example
2	Last digit is even (0, 2, 4, 6, 8)	420
3	Digits add up to a multiple of 3	945: 9 + 4 + 5 = 18, yes
5	Last digit is 0 or 5	945
7	No easy trick: just divide	91 = 7 × 13
10	Last digit is 0	420

Is a number prime? Try dividing by primes 2, 3, 5, 7, 11, ... You only need to go up to the square root of the number. Example: is 51 prime? 5 + 1 = 6, divisible by 3. 51 = 3 × 17. **Not prime.** Is 53 prime? $\sqrt{53} \approx 7.3$. It is not divisible by 2, 3, 5 or 7. **Prime.**

The Fundamental Theorem of Arithmetic

A **theorem** is a statement that has been proved true.

The Fundamental Theorem of Arithmetic says: every integer greater than 1 is either a prime itself, or can be written as a product of primes in **exactly one way** (ignoring the order).

This means primes are the "building blocks" or atoms of all numbers. Breaking a number into primes is called **prime factorisation**.

How to find a prime factorisation

Keep dividing by the smallest prime that works, until you reach 1.

Example 1: 84 (lecture)

Number	Divide by	Result
84	2	42
42	2	21
21	3	7
7	7	1

The primes you divided by: 2, 2, 3, 7. So **$84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$**.

Example 2: 420 (Seminar 1 Task 2.2)

420 ÷ 2 = 210, ÷ 2 = 105, ÷ 3 = 35, ÷ 5 = 7, ÷ 7 = 1. So **$420 = 2^2 \times 3 \times 5 \times 7$**.

In the seminar's story: the storage splits into prime-sized blocks of 2, 2, 3, 5 and 7 units.

Example 3: 945 (Seminar 1 Task 2.3)

945 is odd, so not ÷ 2. Digits add to 18, so ÷ 3: 945 ÷ 3 = 315. 315 ÷ 3 = 105. 105 ÷ 3 = 35. 35 ÷ 5 = 7. 7 ÷ 7 = 1.

So **$945 = 3^3 \times 5 \times 7$**.

Example 4: 30 (Task 2.1) — 30 = **$2 \times 3 \times 5$**.

Check your answer by multiplying back: $2^2 \times 3 \times 5 \times 7 = 4 \times 3 \times 5 \times 7 = 420$. Correct.

Mini tasks 3

1. Is 91 prime? Is 97 prime?
2. Find the prime factorisation of 360.
3. Find the prime factorisation of 1001.
4. Which of these is the correct prime factorisation of 120: (A) 2×60 (B) $2^3 \times 15$ (C) $2^3 \times 3 \times 5$ (D) 4×30?

4. The division theorem

The statement

For any two integers a and b, where $b \neq 0$, there are **unique** integers q and r such that:

```
a = q \cdot b + r \qquad \text{where } 0 \le r < |b|
```

- a is the **dividend** (the number being divided).
- b is the **divisor** (what you divide by).
- q is the **quotient** (how many whole times b fits).
- r is the **remainder** (what's left over).
- **Unique** means there is only one correct pair q and r.
- The condition 0 ⩽ r < |b| means the remainder is **never negative** and is **always smaller than the divisor**.

It is just Phase 0's "divide with remainder", written as a formula. The lecture also writes it as **a / b = q rem r**, and you can find the remainder with **r = a − q × b**.

If r = 0, we say **b divides a**, or a is **divisible** by b.

Worked example (lecture)

a = 17, b = 3. 3 fits into 17 five times (15), leaving 2. So q = 5, r = 2, and 17 = 5 × 3 + 2.

The method for word problems

1. Identify the total amount (a) and the size of one group, block or task (b).
2. Divide: q = the whole number part of a ÷ b. Drop any decimal; do **not** round up.
3. Remainder: r = a − q × b.
4. Answer in words. Read the question carefully: does it want the remainder, or the **empty** space left in a block?

Example 1: CPU scheduling (Seminar 1 Task 3.1)

A CPU has 500 clock cycles. Each task needs 73. How many full tasks, and how many cycles remain?

- a = 500, b = 73.
- 500 ÷ 73 ≈ 6.85, so q = 6. (73 × 6 = 438; 73 × 7 = 511 is too much.)
- r = 500 − 438 = 62.
- **6 full tasks, 62 cycles remain.** Check: 6 × 73 + 62 = 500.

Example 2: File storage (Seminar 1 Task 3.2)

A 900 MB file goes into blocks of 256 MB. How many full blocks, and how much is in the last block?

- 900 ÷ 256 ≈ 3.5, so q = 3 full blocks (768 MB).
- r = 900 − 768 = 132.
- **3 full blocks; the last (4th) block contains 132 MB.**

Example 3: Video on an SSD (revision Task 1)

1750 MB file, blocks of 320 MB.

- 1750 ÷ 320 = 5.47, so q = **5 full blocks** (1600 MB).
- r = 1750 − 1600 = **150 MB** goes into the last block.
- Space left **unused** in the last block = 320 − 150 = **170 MB**.

This is the trap: "how much is **in** the last block" = 150, but "how much space is **left**" = 170. Read the wording.

Negative dividends

The remainder must still be between 0 and |b| − 1. So with a negative a, the quotient goes "one step further" negative.

Example: a = −11, b = 4. If q = −2, then −11 = −8 + r gives r = −3, which is negative, so not allowed. Try q = −3: −11 = −12 + r, so **r = 1**. That works.

This matters for modular arithmetic with negative numbers, next section.

Mini tasks 4

1. Write 47 = q × 6 + r with the correct q and r.
2. A server has 1000 GB. Each backup takes 85 GB. How many full backups fit, and how much space is left over?
3. 2000 log lines are split into pages of 64 lines. How many full pages are there, and how many lines are on the last page?
4. Find q and r for a = −20, b = 7.

5. Modular arithmetic

The idea: arithmetic that wraps around

Think of a 12-hour clock. It is 8 o'clock. In 10 hours it won't be "18 o'clock"; it will be 6 o'clock. After 12, the numbers **wrap around** to the start.

Modular arithmetic is arithmetic where numbers wrap around after reaching a fixed value. That fixed value is the **modulus**. On a clock, the modulus is 12.

The mod operation

a mod b means: **the remainder when a is divided by b.** It is the r from the division theorem.

- 14 mod 5 = 4, because 14 = 2 × 5 + 4.
- 23 mod 7 = 2, because 23 = 3 × 7 + 2.
- 7 mod 2 = 1. 8 mod 4 = 0.
- 3 mod 10 = 3. (If a is smaller than b, the answer is just a.)

The answer to a mod b is **always** one of 0, 1, 2, ..., b − 1. So x mod 7 is always between 0 and 6.

Compare the two notations: the division theorem writes 14/5 = 2 rem 4; modular arithmetic writes 14 mod 5 = 4. Mod keeps **only the remainder** and throws away the quotient.

Congruence (≡)

Two numbers are **congruent modulo n** if they leave the same remainder when divided by n. We write:

```
14 \equiv 4 \pmod 5
```

Read it: "14 is congruent to 4, mod 5." Both leave remainder 4 when divided by 5.

All numbers congruent to 14 (mod 5): {..., −6, −1, 4, 9, 14, 19, 24, ...}. Notice they are spaced exactly 5 apart. **Adding or subtracting the modulus never changes the answer.**

Quick test: a ≡ b (mod n) exactly when a − b is divisible by n. Example: is 23 ≡ 9 (mod 7)? 23 − 9 = 14, divisible by 7. Yes.

Large numbers

Subtract the biggest multiple of the modulus that fits.

- 50 mod 6: 6 × 8 = 48. 50 − 48 = **2**.
- 2570 mod 2048: 2570 − 2048 = **522**.
- 1000 mod 7: 7 × 142 = 994. 1000 − 994 = **6**.

Negative numbers (students lose marks here)

The answer must still be between 0 and n − 1, so it can **never be negative**.

Method: keep adding the modulus until you get a number from 0 to n − 1.

- −5 mod 3: −5 + 3 = −2, still negative. −2 + 3 = **1**.
- −11 mod 4: −11 + 4 = −7, + 4 = −3, + 4 = **1**.
- −20 mod 7: −20 + 7 = −13, + 7 = −6, + 7 = **1**.

Faster method: work out the positive version, then subtract from the modulus. For −20 mod 7: 20 mod 7 = 6, then 7 − 6 = **1**. (If the positive version gives 0, the answer is 0.)

Why? −20 = −3 × 7 + 1. That is the division theorem with q = −3 and r = 1.

Warning: a calculator or some programming languages give −20 mod 7 = −6. In this module, the answer is always the non-negative one: 1.

Even and odd

x mod 2 is 0 for even numbers and 1 for odd numbers. This connects binary and mod: the last binary digit of a number is exactly that number mod 2.

Wrap-around word problems

General method:

1. Find the start position.
2. Add the movement (subtract if moving backwards or deleting).
3. Take the result mod the size of the circle.

Example 1: Clock (lecture). It is 8:00 on a 12-hour clock. What time is it after 22 hours? 8 + 22 = 30. 30 mod 12 = **6**. So 6 o'clock.

Example 2: Ice ring (Seminar 1 Task 4.4). A ring of 360 units. The player is at 40 and moves 240 backwards.

- 40 − 240 = −200.
- −200 mod 360: −200 + 360 = **160**.

Example 3: Circular log buffer (revision Task 2). The buffer holds 2048 entries and has 480. The system deletes 2570.

- 480 − 2570 = −2090.
- −2090 + 2048 = −42. Still negative.
- −42 + 2048 = **2006** entries.

Example 4: Load balancing (Seminar 1 Task 4.3). Requests go to databases using mod 7. Request number 45 goes to 45 mod 7. 7 × 6 = 42, so 45 − 42 = **3**. Database 3.

Example 5: Simple encryption (Seminar 1 Task 4.2). Add 15, then mod 12. The original number is 10. (10 + 15) mod 12 = 25 mod 12 = **1**.

Example 6: Function with mod (Seminar 3 Task 10). $f(x) = (2x + 3) \bmod 7$.

- $f(4) = 11 \bmod 7 =$ **4**
- $f(8) = 19 \bmod 7 =$ **5**
- $f(11) = 25 \bmod 7 =$ **4**
- $f(15) = 33 \bmod 7 =$ **5**
- $f(18) = 39 \bmod 7 =$ **4**

Counting from 1 instead of 0: mock Question 6

The pool has 150 IP addresses, numbered 1 to 150. The server is currently on the 147th. It allocates 10 more addresses. Which number is it on?

Method 1: count by hand. 148, 149, 150 (that's 3), then wrap to 1, 2, 3, 4, 5, 6, 7 (that's 7 more). Total 10. It is on the **7th** address.

Method 2: mod. 147 + 10 = 157. 157 mod 150 = **7**. Answer **B (7th)**.

When things are numbered from 1 (not 0), mod still works, except that a result of 0 means the **last** item. Example: 147 + 3 = 150, and 150 mod 150 = 0, which means the 150th address.

The wrong answers are traps: 157 forgets the wrap; 107 subtracts 50 instead of 150; 8 is an off-by-one error.

Where mod is used in computing (lecture)

Cryptography (RSA, Diffie–Hellman), hash tables, error detection (CRC checks), random number generators, computer graphics (colour wrapping) and scheduling (clocks, calendars).

Mini tasks 5

1. 37 mod 6
2. −7 mod 3
3. 100 mod 12
4. Is 38 ≡ 3 (mod 5)?
5. It is 21:00 on a 24-hour clock. What time is it 50 hours later?
6. A playlist has 12 songs, numbered 1 to 12. You are on song 10 and skip forward 30 songs. Which song are you on?
7. Today is Tuesday. What day is it in 100 days?

6. Ciphers: encrypting words with mod 26

The words

- **Encryption** turns a readable message into a secret one.
- **Decryption** turns it back.
- The readable message is the **plaintext** (P). The secret one is the **ciphertext** (C).
- A **cipher** is the method (the algorithm) used to encrypt and decrypt.
- The **key** (k) is the secret number used by the cipher.

Letters as numbers

The alphabet has 26 letters, so we use **mod 26**. Number the letters **starting from 0**:

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

In the exam you can't carry this table, so write it out at the start. Landmarks to remember: **a = 0, e = 4, j = 9, o = 14, t = 19, z = 25**.

Check whether the question starts at 0 or 1. Your lecture and mock both start at a = 0.

The Caesar cipher

Shift every letter forward by k places, wrapping round from z to a.

```
\text{Encrypt: } C \equiv P + k \pmod{26} \qquad \text{Decrypt: } P \equiv C - k \pmod{26}
```

Example: decrypt "UPQY" with k = 2 (lecture homework).

Cipher letter	U	P	Q	Y
Number C	20	15	16	24
C − 2	18	13	14	22
Plain letter	S	N	O	W

The message is **SNOW**.

Wrap-around example: encrypt "y" with k = 3. y = 24. 24 + 3 = 27. 27 mod 26 = 1 = **b**. Decrypting b with k = 3: 1 − 3 = −2, and −2 + 26 = 24 = **y**.

Multiply-and-add ciphers (mock Question 23)

Some ciphers multiply first, then add, then take mod 26:

```
C = (a \cdot P + b) \bmod 26
```

Mock Q23: multiply each letter by 4, add 1, take mod 26. Encrypt "dog".

Letter	d	o	g
P	3	14	6

Letter	d	o	g
4P + 1	13	57	25
mod 26	13	57 − 52 = 5	25
Cipher letter	n	f	z

"dog" encrypts to **nfz**. Answer **C**.

Exam shortcut: work out only the first letter, then eliminate. d gives n, so only options starting with n can be right. Here only C (nfz) does.

Reverse problem: which input gives this output? (Seminar 1 Task 4.5, challenge)

The formula is y = 4x + 5, then the code is y mod 11. The code is 8, and x is a whole number from 0 to 9. What is x?

Method: just try every x. There are only ten.

x	0	1	2	3	4	5	6	7	8	9
4x + 5	5	9	13	17	21	25	29	33	37	41
mod 11	5	9	2	6	10	3	7	0	4	8

Only **x = 9** gives 8. (41 = 3 × 11 + 8.)

Mini tasks 6

1. Encrypt "cab" with the Caesar cipher, k = 3.
2. Decrypt "KHOOR" with the Caesar cipher, k = 3.
3. Encrypt "cat" with C = (3P + 2) mod 26.
4. Encrypt "zoo" with C = (4P + 1) mod 26.

7. Sequences

The words

- A **sequence** is an **ordered list** of numbers (or objects) that follows a rule. Order matters, and numbers may repeat.
- Each item is a **term**. Its position is its **index**. In (2, 4, 6, 8), the term at index 3 is 6.

- We write the nth term as $\mathbf{a_n}$ ("a sub n"). So a_1 is the first term, a_2 the second, and so on.
- A sequence can be **finite** (it stops) or **infinite** (it goes on forever, shown by "...").

Arithmetic sequences: add the same number each time

An **arithmetic sequence** (or arithmetic progression) adds the same number each step. That number is the **common difference**, d.

- 2, 5, 8, 11, ... has d = 3.
- 10, 7, 4, 1, ... has d = −3 (it goes down).
- To find d: **d = any term minus the term before it.**

General form: a, a + d, a + 2d, a + 3d, ... where a is the first term.

```
a_n = a + (n - 1)\,d
```

Why n − 1? To reach the nth term you take n − 1 steps from the first term. The 15th term is 14 steps after the 1st.

Example 1 (Seminar 1 Task 5.1): find the 15th term of 3, 8, 13, 18, ...

- a = 3, d = 8 − 3 = 5, n = 15.
- a_{15} = 3 + (15 − 1) × 5 = 3 + 14 × 5 = 3 + 70 = **73**.

Example 2 (Task 5.6): find the formula for the nth term of 10, 7, 4, 1, ...

- a = 10, d = −3.
- a_n = 10 + (n − 1)(−3) = 10 − 3n + 3 = **13 − 3n**.
- Check: n = 1 gives 10; n = 4 gives 1. Correct.

Example 3 (working backwards): in 4, 10, 16, 22, ..., which term equals 100?

- a = 4, d = 6. Solve 4 + (n − 1) × 6 = 100.
- (n − 1) × 6 = 96, so n − 1 = 16, so **n = 17**.

Geometric sequences: multiply by the same number each time

A **geometric sequence** (or geometric progression) multiplies by the same number each step. That number is the **common ratio**, r. r can't be 0.

- 3, 6, 12, 24, ... has r = 2.
- 81, 27, 9, 3, ... has r = 1/3 (it shrinks).
- To find r: **r = any term divided by the term before it.**

General form: a, ar, ar^2, ar^3, ...

$$a_n = a \cdot r^{\,n-1}$$

Again n − 1, because the first term has not been multiplied yet.

Example 4 (revision Task 3): 5, 20, 80, 320, ... Find its type and the 6th term.

- Ratios: 20 ÷ 5 = 4, 80 ÷ 20 = 4, 320 ÷ 80 = 4. Geometric, r = 4.
- $a_6 = 5 \times 4^5 = 5 \times 1024 =$ **5120**.

Example 5 (Task 5.2): first term 5, ratio 3. Find the 4th term. $a_4 = 5 \times 3^3 = 5 \times 27 =$ **135**.

Example 6 (Task 5.5): 7, 21, 63, 189, ... Ratio 3. $a_5 = 7 \times 3^4 = 7 \times 81 =$ **567**. (Or just continue: 189 × 3 = 567.)

How to identify the type

1. Work out the **differences** between neighbouring terms. All the same? **Arithmetic.**
2. If not, work out the **ratios**. All the same? **Geometric.**
3. Neither? Then it is **neither** (or something else, like Fibonacci or squares).

Sequence	Differences	Ratios	Type
4, 10, 16, 22	6, 6, 6	—	Arithmetic, d = 6
6, 18, 54, 162	12, 36, 108	3, 3, 3	Geometric, r = 3
1, 4, 9, 16	3, 5, 7	4, 2.25, ...	Neither (square numbers)
2, 3, 5, 8, 13	1, 2, 3, 5	not constant	Neither (Fibonacci-type)

The Fibonacci sequence

In the **Fibonacci sequence**, each term is the **sum of the two terms before it**. It starts 0, 1:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

As a rule:

$$F(n) = F(n-1) + F(n-2), \qquad F(0) = 0,\; F(1) = 1$$

A rule that defines each term using earlier terms is called a **recurrence relation**. The starting values are the **initial conditions**. Phase 3 calls these "base cases" and "recursive cases"; this is your first taste.

Golden ratio: divide each Fibonacci number by the one before it (8/5 = 1.6, 13/8 = 1.625, 21/13 ≈ 1.615, ...). These ratios get closer and closer to **φ ≈ 1.618**, the golden (or divine) ratio, found in art, architecture and nature.

Modified Fibonacci (Seminar 1 Task 5.7)

Same rule, different start. Blocks are 3 MB and 5 MB, then each is the sum of the previous two. Find the 7th block.

Block	1st	2nd	3rd	4th	5th	6th	7th
Size (MB)	3	5	8	13	21	34	**55**

Careful with counting: decide whether the first term is called term 0 or term 1. Here the question says "first block", so 3 is the 1st.

Mini tasks 7

1. Find the 10th term of 2, 9, 16, 23, ...
2. Identify 81, 27, 9, 3, ... and give the next term.
3. Identify 2, 4, 8, 14, 22, ...
4. Find the formula for the nth term of 5, 9, 13, 17, ...
5. A geometric sequence starts 2, 6, 18, ... Which term equals 486?
6. A Fibonacci-style sequence starts 2, 3. What is its 7th term?

Phase 1 practice questions (exam style)

These are built like your real test, and some are harder than the mock. Give yourself about 30 minutes, no calculator.

1. What is 200 in binary? (A) 11001000 (B) 11000100 (C) 10101000 (D) 11001001
2. What is 1111111_2 in decimal? (A) 255 (B) 127 (C) 128 (D) 63
3. Which binary number equals 90? (A) 1011010 (B) 1011011 (C) 1101010 (D) 101101
4. Which of these is **not** a prime number? (A) 2 (B) 29 (C) 51 (D) 53

5. The prime factorisation of 360 is: (A) $2^3 \times 3^2 \times 5$ (B) $2^2 \times 3^3 \times 5$ (C) $2^3 \times 45$ (D) $2 \times 3 \times 5 \times 12$
6. What is $-17 \bmod 5$? (A) −2 (B) 2 (C) 3 (D) −3
7. Which number is **not** congruent to 14 (mod 5)? (A) −1 (B) 19 (C) 34 (D) 21
8. A CPU has 1000 clock cycles. Each task needs 85 cycles. How many full tasks run, and how many cycles remain? (A) 11 tasks, 65 left (B) 12 tasks, 20 left (C) 11 tasks, 85 left (D) 11 tasks, 20 left
9. It is 22:00 on a 24-hour clock. What time is it 30 hours later? (A) 52:00 (B) 06:00 (C) 04:00 (D) 08:00
10. A circular buffer has 60 slots, numbered 0 to 59. A pointer at slot 45 moves 100 slots backwards. Where does it land? (A) 5 (B) 55 (C) 15 (D) −55
11. Letters are numbered a = 0 to z = 25. Each letter is encrypted with $C = (3P + 2) \bmod 26$. How is "cat" encrypted? (A) icg (B) ich (C) hcb (D) jdi
12. A Caesar cipher with key k = 5 produced "FUUQJ". What was the original word?
13. What is the 20th term of the arithmetic sequence 7, 11, 15, 19, ...? (A) 83 (B) 87 (C) 80 (D) 79
14. What is the 8th term of the geometric sequence 3, 6, 12, 24, ...? (A) 768 (B) 384 (C) 192 (D) 48
15. The sequence 1, 4, 9, 16, 25, ... is: (A) arithmetic (B) geometric (C) Fibonacci (D) neither arithmetic nor geometric
16. A sequence starts 2, 5 and each next term is the sum of the previous two. What is the 6th term? (A) 19 (B) 31 (C) 50 (D) 26
17. Which of these is irrational? (A) 0.75 (B) −3 (C) $\sqrt{16}$ (D) $\sqrt{5}$
18. 1000 users must be hosted on servers that each hold 32 users. How many servers are needed to host **everyone**? (A) 31 (B) 32 (C) 30 (D) 33
19. A code is computed as $(3x + 1) \bmod 7$. The code is 5, and x is a whole number from 0 to 6. What is x? (A) 2 (B) 4 (C) 5 (D) 6
20. What is the nth-term formula of 20, 17, 14, 11, ...? (A) $20 - 3n$ (B) $23 - 3n$ (C) $17 + 3n$ (D) $3n + 20$

Answers with full explanations

Mini tasks 1 (binary)

1. $53 = 32 + 16 + 4 + 1$. Positions 32, 16, 8, 4, 2, 1 → 1, 1, 0, 1, 0, 1 = **110101**.

2. 102 = 64 + 32 + 4 + 2. Positions 64 to 1 → **1100110**.
3. 111000: 32 + 16 + 8 = **56**.
4. 11111: 16 + 8 + 4 + 2 + 1 = **31**.
5. 38 = 32 + 4 + 2 → 100110. **(A).** Shortcut: 38 is even, so (B) is out; it needs 6 digits, so (D) is out.

Mini tasks 2 (types of numbers)

1. 12: natural. −3: integer. 2/5: rational. √7: irrational. √49 = 7: natural. 0: whole number (natural only if your ℕ includes 0).
2. **True.** Any integer n can be written n/1.
3. **False.** √2 can't be written as a fraction; it is irrational.
4. **float** (or double).

Mini tasks 3 (primes)

1. 91 = 7 × 13, **not prime**. 97: not divisible by 2, 3, 5 or 7, and √97 < 10, so **97 is prime**.
2. 360 ÷ 2 = 180, ÷ 2 = 90, ÷ 2 = 45, ÷ 3 = 15, ÷ 3 = 5, ÷ 5 = 1. **$360 = 2^3 \times 3^2 \times 5$**.
3. 1001 ÷ 7 = 143, ÷ 11 = 13. **$1001 = 7 \times 11 \times 13$**.
4. **(C).** The others contain non-primes (60, 15, 4, 30). A prime factorisation may only contain primes.

Mini tasks 4 (division theorem)

1. 6 × 7 = 42, 47 − 42 = 5. **$47 = 7 \times 6 + 5$**.
2. 85 × 11 = 935; 85 × 12 = 1020 is too much. **11 backups, 1000 − 935 = 65 GB left.**
3. 64 × 31 = 1984. **31 full pages; the last page has 2000 − 1984 = 16 lines.**
4. q = −3, r = 1, because −20 = (−3) × 7 + 1 and $0 \leqslant 1 < 7$.

Mini tasks 5 (mod)

1. 6 × 6 = 36. 37 mod 6 = **1**.
2. −7 + 3 = −4, + 3 = −1, + 3 = **2**.
3. 12 × 8 = 96. 100 mod 12 = **4**.
4. 38 − 3 = 35, which is divisible by 5. **Yes.** (Both leave remainder 3.)
5. 21 + 50 = 71. 71 mod 24 = 71 − 48 = **23, so 23:00**.

6. 10 + 30 = 40. 40 mod 12 = 4. **Song 4.** (Count check: 30 = 2 full loops of 12 + 6 more; 10 + 6 = 16, wrap to 4.)
7. 100 mod 7 = 2 (7 × 14 = 98). Two days after Tuesday is **Thursday**.

Mini tasks 6 (ciphers)

1. c = 2 → 5 = f. a = 0 → 3 = d. b = 1 → 4 = e. **"fde"**.
2. K = 10 → 7 = h. H = 7 → 4 = e. O = 14 → 11 = l. O → l. R = 17 → 14 = o. **"hello"**.
3. c = 2 → 3 × 2 + 2 = 8 = i. a = 0 → 2 = c. t = 19 → 59; 59 − 52 = 7 = h. **"ich"**.
4. z = 25 → 4 × 25 + 1 = 101; 101 − 78 = 23 = x. o = 14 → 57; 57 − 52 = 5 = f. **"xff"**.

Mini tasks 7 (sequences)

1. d = 7. a_{10} = 2 + 9 × 7 = **65**.
2. Ratio 27 ÷ 81 = 1/3. **Geometric**, r = 1/3. Next term 3 × 1/3 = **1**.
3. Differences 2, 4, 6, 8 are not equal; ratios 2, 2, 1.75 are not equal. **Neither.** (The differences grow by 2, so the next term is 32.)
4. a = 5, d = 4. a_n = 5 + (n − 1) × 4 = **4n + 1**.
5. 2 × 3^{n-1} = 486, so 3^{n-1} = 243 = 3^5. So n − 1 = 5, **n = 6**. (Check by listing: 2, 6, 18, 54, 162, 486.)
6. 2, 3, 5, 8, 13, 21, **34**.

Phase 1 practice questions

1. **(A)** 200 = 128 + 64 + 8. Positions 128, 64, 32, 16, 8, 4, 2, 1 → 11001000.
2. **(B)** 64 + 32 + 16 + 8 + 4 + 2 + 1 = 127. Shortcut: n ones in a row = 2^n − 1, so seven ones = 128 − 1.
3. **(A)** 90 = 64 + 16 + 8 + 2 → 1011010. (B) is odd, so it can't be 90.
4. **(C)** 5 + 1 = 6, divisible by 3. 51 = 3 × 17.
5. **(A)** 360 = 2^3 × 3^2 × 5 = 8 × 9 × 5 = 360. (C) and (D) contain non-primes; (B) = 4 × 27 × 5 = 540.
6. **(C)** −17 + 5 + 5 + 5 + 5 = 3. Check: −17 = (−4) × 5 + 3. Mod is never negative, so (A) and (D) are out.
7. **(D)** 14 mod 5 = 4. −1 mod 5 = 4, 19 mod 5 = 4, 34 mod 5 = 4, but 21 mod 5 = **1**.
8. **(A)** 85 × 11 = 935. 1000 − 935 = 65.
9. **(C)** 22 + 30 = 52. 52 mod 24 = 52 − 48 = 4. So 04:00.

10. **(A)** 45 − 100 = −55. −55 + 60 = 5.
11. **(B)** c → 3 × 2 + 2 = 8 = i; a → 2 = c; t → 59 mod 26 = 7 = h. Shortcut: only (A) and (B) start "ic", so you only need the last letter.
12. **"apple"**. F = 5 → 0 = a. U = 20 → 15 = p. U → p. Q = 16 → 11 = l. J = 9 → 4 = e.
13. **(A)** a = 7, d = 4. a_{20} = 7 + 19 × 4 = 7 + 76 = 83.
14. **(B)** a = 3, r = 2. $a_8 = 3 \times 2^7 = 3 \times 128 = 384$. (A) uses 2^8, a classic n instead of n − 1 error.
15. **(D)** Differences 3, 5, 7 and ratios 4, 2.25 are not constant. These are the square numbers.
16. **(B)** 2, 5, 7, 12, 19, **31**.
17. **(D)** $\sqrt{16} = 4$ is natural. 0.75 = 3/4 is rational. $\sqrt{5}$ has no exact fraction form.
18. **(B)** 1000 = 31 × 32 + 8. 31 servers are full, but 8 users still need a server. So **32**. The question asked for everyone, not for full servers.
19. **(D)** Try each x: 3x + 1 gives 1, 4, 7, 10, 13, 16, 19. Mod 7: 1, 4, 0, 3, 6, 2, 5. Only x = 6 gives 5. Check: 19 = 2 × 7 + 5.
20. **(B)** a = 20, d = −3. a_n = 20 + (n − 1)(−3) = 20 − 3n + 3 = 23 − 3n. Check: n = 1 gives 20.

Scoring: 16 or more right means you're solid on Phase 1. If you missed a mod or binary question, redo those mini tasks before moving on.

When you're ready, ask for **Phase 2: Sets and intervals**.